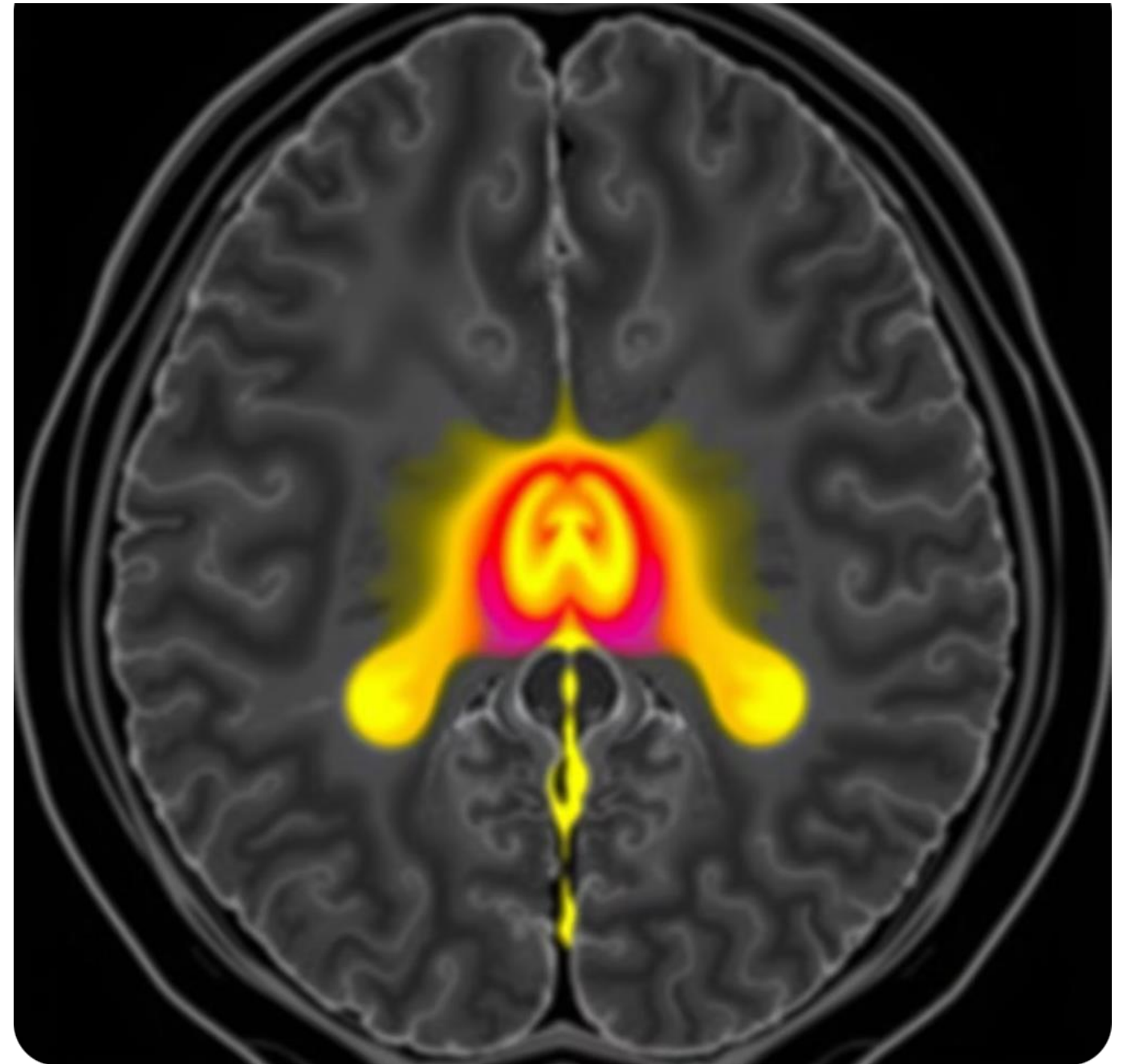


BRAIN TUMOR DETECTION SYSTEM

- This AI-powered system uses deep learning to analyze MRI scans and detect brain tumors. It offers a user-friendly interface for uploading images, instant analysis, and downloadable PDF reports. Developed with advanced technologies, it aims to improve early and accurate diagnosis, reducing manual inspection errors and time.
-
- Presented by: Livingstone Mazvovere
- Exam Roll No: 22020107055



Project Overview and Timeline

- 1 Phase 1: Requirement Analysis and Data Collection
Week 1 focused on gathering necessary data and defining project requirements.
- 2 Phase 2 to 6: Development to Delivery
Weeks 2-6 covered data preparation, model development, UI design, deployment, and final testing.

Technologies and Setup

Core Technologies

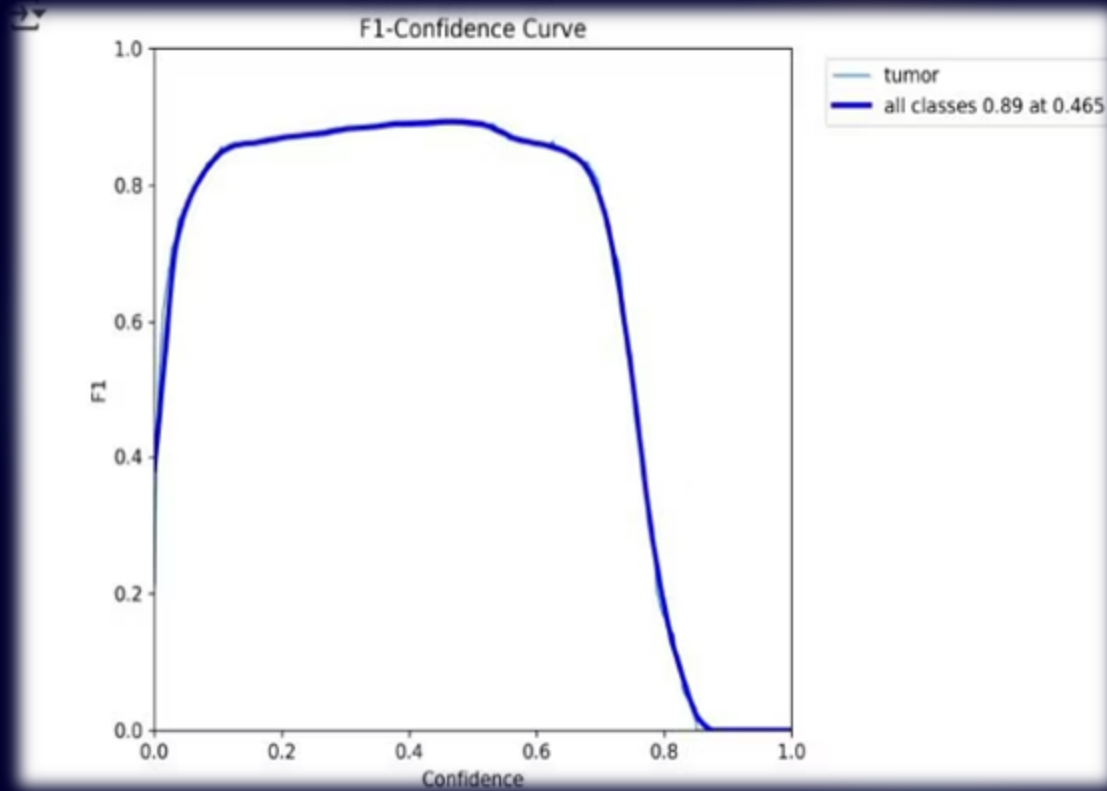
- Streamlit for web interface
- OpenCV for image processing
- YOLO (Ultralytics) for tumor detection
- FPDF for PDF report generation

Installation Steps

- Clone repository from GitHub
- Create virtual environment
- Install dependencies via pip
- Run the app using Streamlit



Model Training and Performance



Training Details

YOLOv11m model fine-tuned on 1500+ MRI scans over 20 epochs, achieving high accuracy and efficiency.

- 125 layers, 20M parameters
- Training time: 0.668 hours
- Inference time: 24.9ms per image

Performance Metrics

- mAP@50: 92.1%
- Precision: 91.4%
- Recall: 87.4%
- Model weights size: 40.5MB

Data Input Module

Handles MRI scan uploads and preprocessing.

Detection Engine

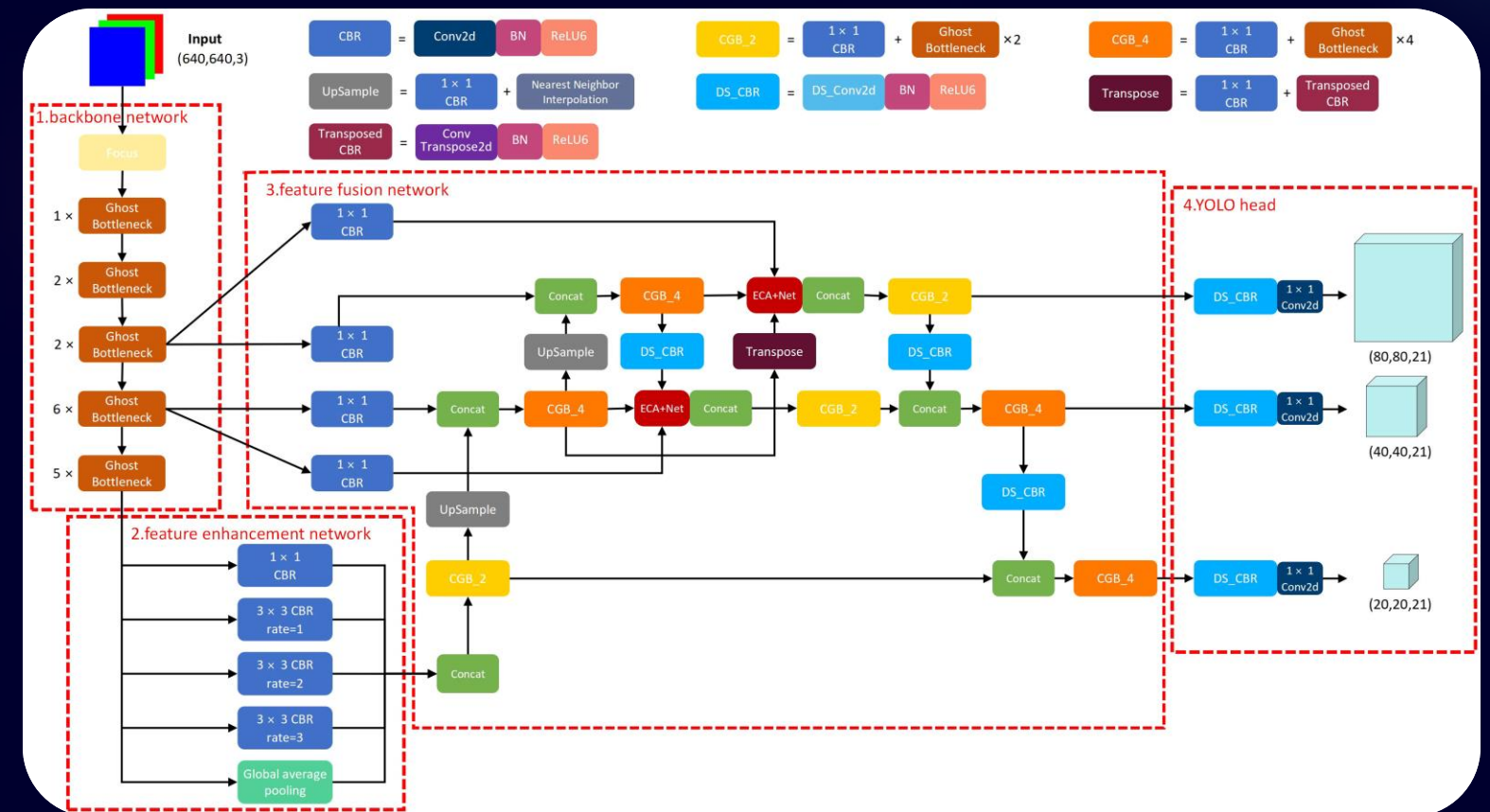
Performs tumor detection using the YOLOv11m model.

Visualization Module

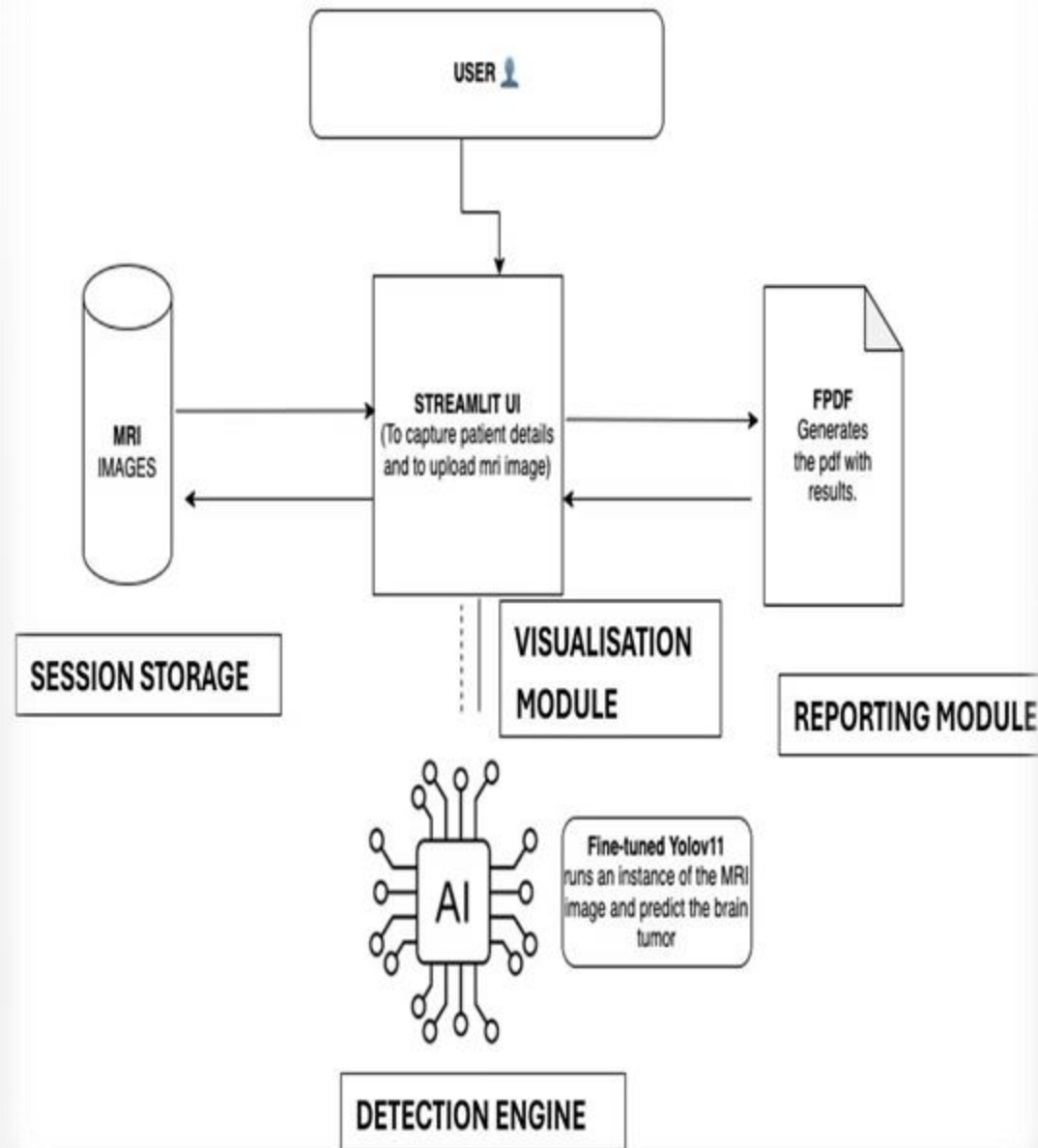
Highlights detected tumors with bounding boxes on images.

Reporting Module

Generates downloadable PDF diagnostic reports.



User Guide and Technical Implementation



User Steps

- 1
 - Enter patient details
 - Upload MRI scan
 - View tumor detection results
 - Download report

Technical Details

- 2

Image preprocessing includes resizing and normalization. Detection uses YOLOv11m. Streamlit renders UI and FPDF generates reports.

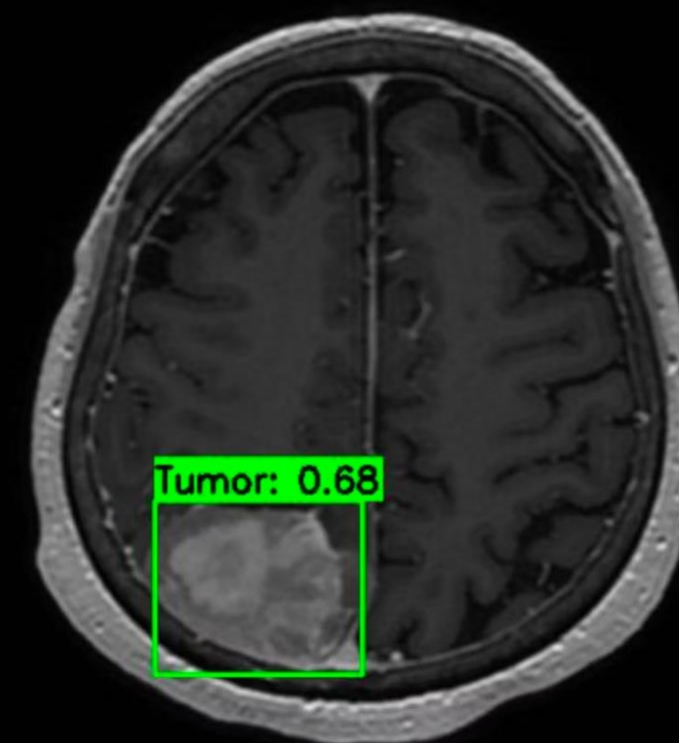
Detection Results and Output

Report Highlights

Includes patient info, tumor locations, confidence scores, and classification (benign/malignant).

Disclaimer

AI-generated reports assist diagnosis but require review by licensed medical professionals.



Conclusion and Future Enhancements

1 Conclusion

Deep learning enables accurate, real-time brain tumor detection, supporting early diagnosis and aiding radiologists.

2 Future Enhancements

- 3D MRI support
- Attention mechanisms
- Mobile app version
- DICOM server integration
- Expert feedback tools